

Commercial Auto Risk Scoring Case Study

Thank you for your interest in data science team at Zurich. As part of our interview process, we have prepared a technical case study for you to complete.

We trust that you will complete this on your own and keep it confidential. Use of AI-assisted development tools is permitted, but please disclose which tools you used and how you used them.

Thank you for the time and effort you invest in this case study — we look forward to reviewing your work.

Background

Commercial auto insurance covers fleets of vehicles operated by businesses. Insurers assess the risk of each fleet to set appropriate premiums, and accurate risk scoring is central to underwriting profitability. It has two main coverages. **Auto Physical Damage (APD)** covers damage to the insured's own vehicles and **Auto Liability (AL)** covers third-party bodily injury and property damage. Policies may carry one or both coverage lines, and the key risk drivers may differ between them.

You are provided with a dataset of approximately 15,000 commercial auto policies from 2020–2022, along with a prototype risk scoring model. An additional 3,000 policies from 2023–2024 are provided for scoring, with no outcomes available for this period.

Included Files

1. **train.csv** — approximately 15,000 labeled policies from 2020–2022 with claim outcomes for model training and validation.
2. **score.csv** — approximately 3,000 unlabeled policies from 2023–2024 for generating predictions.
3. **data_dictionary.md** — definitions for all columns.
4. **risk_scorer.ipynb** — **prototype Jupyter notebook and starting point for your work.**

Data Dictionary Preview

A selection of key columns is shown below. Full definitions are provided in **data_dictionary.md**.

policy_id	Unique policy identifier
coverage_type	APD (Auto Physical Damage) or AL (Auto Liability). An insured may have both.
vehicle_count	Number of vehicles in the insured's fleet
driver_count	Number of drivers listed on the policy
prior_apd_claim_count	Number of APD claims in the prior policy period
prior_loss_amount	Total prior period loss amount in USD
claim_count	Number of claims in the current period
total_loss_amount	Total loss amount in the current period

The Case Study Tasks

Your main deliverable is a proposal for modeling methodology and production readiness roadmap.

You are **NOT** expected to deliver model predictions or end-to-end solutions, although you are free to include any Python code or pseudocode in the provided repository. The expected time commitment is 4-6 hours.

Your analysis / methodology proposal should include the following:

1. **Exploratory data analysis.**

How would you explore the dataset and what patterns could influence modeling choices?

2. **Modeling methodology improvements.**

Are there any impactful changes or additions to the modeling methodology that could improve predictive performance on unseen data? Limit to top three improvements.

3. **Production readiness improvements.**

Are there any impactful changes or additions that could increase the current production readiness of the solution? Limit to top three improvements.

4. **Deployment roadmap.**

Briefly outline further steps that would need to be completed before this model could be deployed to production.

5. **Technical presentation.**

Prepare a 15–20 minute **English** presentation covering your **methodology proposal** on exploration, modeling methodology, production readiness and deployment roadmap. You can also mention new concepts learned while working on this case study, as well as disclose AI-assisted development tools used.

AI tools: Use of AI development tools is permitted. Please disclose in presentation which tools you used and how you used them.

Sharing Your Results

Please commit all work — presentation, notes and notebooks / codes (if any) — to a **public GitHub repository** and send the link to Milan Polak (milan.polak@zurich.com) and Daniel Gabris (daniel.gabris@zurich.com) **within the deadline provided by HR or managers.**

Next steps

Your submission will be reviewed by the team. If it passes the review, you will be invited to a follow-up interview where you will have the opportunity to present your findings and discuss them with hiring managers and invited Bratislava team members. **This interview will be fully conducted in English.**